

KAZALY, Kazakhstan

WMO: 358490

Lat: 45.7556N

Lon: 62.1067E

Elev: 223

StdP: 14.58

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-7.4	-19.8	1.9	-11.3	-14.4	2.5	-5.9	27.4	15.2	22.6	20.4	3.5	50	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	99.3	69.0	96.5	68.0	93.9	66.9	71.3	93.6	69.8	92.2	68.4	89.7	4.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	89.5	80.5	61.9	83.5	79.4	60.0	78.1	78.9	35.3	93.4	33.9	92.4	32.8	89.7	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.5	16.2	13.5	-17.3	104.1	7.0	3.2	-22.4	106.4	-26.4	108.2	-30.4	110.0	-35.4	112.3		
			-17.7	74.6	6.9	2.9	-22.6	76.7	-26.7	78.4	-30.5	80.1	-35.5	82.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	15.2	18.9	37.2	54.3	67.8	76.7	79.8	76.7	64.2	48.7	33.7	20.6	
	DBStd	24.78	12.03	13.04	11.11	9.92	7.35	6.34	5.67	6.78	7.97	8.63	9.94	10.92	
	HDD50	3975	1078	872	414	68	1	0	0	0	4	131	495	912	
	HDD65	7032	1543	1292	862	340	55	4	0	5	107	508	939	1377	
	CDD50	3847	0	0	18	198	554	800	923	828	431	90	5	0	
	CDD65	1429	0	0	0	19	143	354	458	369	84	2	0	0	
	CDH74	18072	0	0	10	267	1751	4584	5967	4438	1006	48	1	0	
	CDH80	9075	0	0	2	82	675	2371	3250	2319	371	5	0	0	
Wind	WSAvg	4.7	4.5	5.0	5.7	5.4	5.9	4.7	4.3	4.3	4.0	4.3	3.9	4.5	
Precipitation	PrecAvg	5.1	0.5	0.4	0.6	0.7	0.5	0.3	0.2	0.2	0.4	0.5	0.6	0.5	
	PrecMax	7.9	1.5	1.3	2.5	3.5	1.7	1.0	1.3	1.1	2.3	2.1	1.7	2.0	
	PrecMin	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	PrecStd	1.5	0.4	0.3	0.4	0.7	0.5	0.3	0.3	0.3	0.5	0.5	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.7	52.0	71.9	88.1	94.8	101.7	103.8	101.7	94.9	79.8	64.3	44.2	
		MCWB	37.3	44.5	54.1	63.2	64.6	69.1	70.9	69.4	64.7	57.6	52.4	37.9	
	2%	DB	36.2	42.9	63.8	81.5	90.6	97.9	99.6	97.9	88.4	74.1	56.3	40.1	
		MCWB	34.4	38.4	50.7	59.9	63.0	68.0	69.6	68.3	63.2	54.3	47.4	37.0	
	5%	DB	33.8	38.4	58.7	75.9	87.2	94.8	96.6	95.0	84.1	69.5	51.6	36.8	
		MCWB	32.0	34.6	47.7	56.7	61.8	67.1	69.0	67.3	61.3	52.7	44.5	34.1	
	10%	DB	31.6	34.8	53.1	71.3	83.5	91.9	93.7	91.8	79.9	64.5	47.1	33.9	
		MCWB	29.8	32.1	44.5	54.5	60.7	65.9	67.7	66.0	59.6	50.5	41.4	31.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.0	46.4	55.5	65.6	67.5	73.1	73.8	73.0	67.3	59.1	53.5	40.1
			MCDB	40.0	50.9	70.1	83.5	87.5	94.9	95.3	95.9	89.9	74.0	61.7	42.1
2%		WB	34.6	39.4	51.3	61.1	65.2	70.5	71.6	70.1	64.4	56.2	48.7	37.1	
		MCDB	36.0	43.3	61.9	76.4	84.9	92.5	94.3	92.8	86.4	71.0	55.0	39.8	
5%		WB	32.2	35.1	48.4	58.2	63.5	68.6	70.2	68.4	62.2	53.8	45.0	34.3	
		MCDB	33.6	37.9	57.4	73.6	82.8	90.0	92.5	91.1	81.3	66.5	50.7	36.6	
10%		WB	29.9	32.4	45.1	55.6	61.9	67.0	69.0	67.1	60.1	51.4	42.0	31.8	
		MCDB	31.5	34.6	52.8	69.1	80.4	87.7	90.4	88.7	78.1	63.3	46.7	33.8	
Mean Daily Temperature Range		5% DB	MDBR	13.3	16.1	18.6	21.6	23.6	23.9	23.3	24.2	24.7	23.1	16.4	13.0
			MCDBR	10.7	17.1	26.1	27.9	27.7	27.2	26.8	27.9	29.1	29.4	21.8	14.3
	MCWBR		9.6	12.6	15.1	12.5	10.0	9.8	9.5	10.5	12.7	15.4	14.1	11.5	
	5% WB	MCDBR	9.9	15.9	24.2	25.4	24.9	24.5	24.5	25.5	27.3	26.3	20.1	13.3	
		MCWBR	9.1	12.3	15.1	12.3	10.5	9.9	9.6	10.5	12.6	15.0	13.6	11.1	
Clear-Sky Solar Irradiance	taub	0.335	0.361	0.430	0.468	0.466	0.446	0.428	0.414	0.380	0.382	0.363	0.342		
	taud	2.083	2.049	1.982	1.929	1.998	2.117	2.193	2.208	2.246	2.154	2.145	2.103		
	Ebn at Noon	229	249	247	250	256	262	265	264	263	239	218	211		
	Edh at Noon	31	39	49	56	54	49	44	42	37	35	30	28		
All-Sky Solar Radiation	RadAvg	402	721	1143	1645	2113	2299	2218	1997	1544	973	534	326		
	RadStd	74	128	100	115	113	123	94	85	57	60	53	53		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-1.31	N/A	N/A	+263	+229
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.46	N/A	N/A	N/A	N/A

Nomenclature: See separate page